Harsh Tripathi

Software Developer | Java & DSA | MERN Stack | Generative AI
Lucknow, India | +91 8858935948 | harshx2204@gmail.com
LinkedIn: linkedin.com/in/harshtripathi2201 | LeetCode: leetcode.com/u/Harsh_0204

PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science & Engineering student at DSMNRU, Lucknow, with hands-on experience building full-stack applications: a Stock Trading Platform, Video Conferencing System, Version Control System, Social Media Platform, and an AI chatbot using the OpenAI API. Proficient in Java, Data Structures & Algorithms, and the MERN Stack (MongoDB, Express.js, React.js, Node.js). Certified in DSA (Java) and MERN Stack by Apna College, and in Generative AI by Outskill and TCS iON. Seeking a Software Development Engineer (SDE) internship or entry-level role.

EDUCATION

Dr. Shakuntala Misra National Rehabilitation University (DSMNRU), Lucknow, India
Bachelor of Technology (B.Tech) in Computer Science & Engineering | 2023 - 2027 | CGPA: 8.39/10

TECHNICAL SKILLS

Programming Languages: Java, JavaScript (ES6+), HTML5, CSS3
Frontend: React.js, Context API, Bootstrap
Backend & Database: Node.js, Express.js, RESTful APIs, WebSocket, Socket.io, JWT Authentication, MongoDB
Real-Time & AI: WebRTC, OpenAI API, Prompt Engineering
Core Computer Science: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Tools: Git, GitHub, VS Code, Postman, npm
Soft Skills: Communication, Problem Solving, Time Management, Decision Making | **Languages:** English, Hindi

PROJECTS

Version Control System (like Git) - Developer | Mar 2025 - Apr 2025
- Built a command-line Version Control System from scratch in Java, replicating core Git functionality: init, add, commit, log, diff, and branch management.
- Implemented content-addressable storage with SHA-based hashing and a directed acyclic graph (DAG) of commits; designed Tree, Blob, and Commit objects using OOP principles.
- Developed branching and merging logic with conflict detection to support parallel development workflows.

Technologies: Java, OOP, File I/O, Data Structures, Hashing

Video Conferencing Platform - Developer | Apr 2025 - May 2025
- Developed a real-time video conferencing application with peer-to-peer video and audio using the WebRTC protocol.
- Implemented a WebSocket signaling server with Socket.io for room creation, user joining, and connection negotiation; handled join, leave, and disconnect events.
- Built multi-participant rooms with mute/unmute, screen sharing, and live chat in a responsive React.js UI for desktop and mobile browsers.

Technologies: React.js, Node.js, Express.js, WebRTC, Socket.io

Stock Trading Platform - Developer | Jul 2025 - Aug 2025
- Developed a full-stack stock trading platform where users can register, browse live listings, place buy/sell orders, and track portfolio performance.
- Built a secure RESTful backend with Node.js and Express.js for order management, transaction history, and JWT-based authentication; integrated external stock market APIs for live pricing.
- Designed a React.js frontend with real-time price updates and interactive dashboards; implemented MongoDB schemas for Users, Stocks, Orders, and Transactions.

Technologies: React.js, Node.js, Express.js, MongoDB, REST APIs, JWT

SigmaGPT - AI Chatbot using OpenAI APIs - Developer | Oct 2025 - Nov 2025
- Built a full-stack ChatGPT-like application using OpenAI's GPT APIs, supporting context-aware multi-turn conversations with streaming, token-by-token responses.
- Developed a Node.js/Express.js backend to securely manage API keys, proxy requests to OpenAI, and persist chat history in MongoDB.
- Designed a React.js chat UI with message history, session management, and markdown rendering; applied prompt engineering to customize AI behavior and personas.

Technologies: React.js, Node.js, Express.js, MongoDB, OpenAI API, Prompt Engineering

Social Media Platform (LinkedIn-like) - Developer | Nov 2025 - Dec 2025
- Built a full-stack social media platform with registration, profiles, posts, likes, comments, and a follow/follower system.
- Implemented authentication and authorization with JWT tokens and bcrypt password hashing; developed RESTful Express.js endpoints for the feed, post CRUD, connections, and notifications.
- Built a React.js frontend with Context API for global state; modeled MongoDB collections for Users, Posts, Comments, and Connections with indexing for efficient queries.

Technologies: React.js, Node.js, Express.js, MongoDB, JWT, bcrypt, Context API

CERTIFICATIONS

- Data Structures & Algorithms using Java - Apna College, 2024
- Full Stack Web Development (MERN Stack) - Apna College, 2025
- Generative AI Certification - Outskill, 2025
- AI For All Programme - TCS iON, 2026